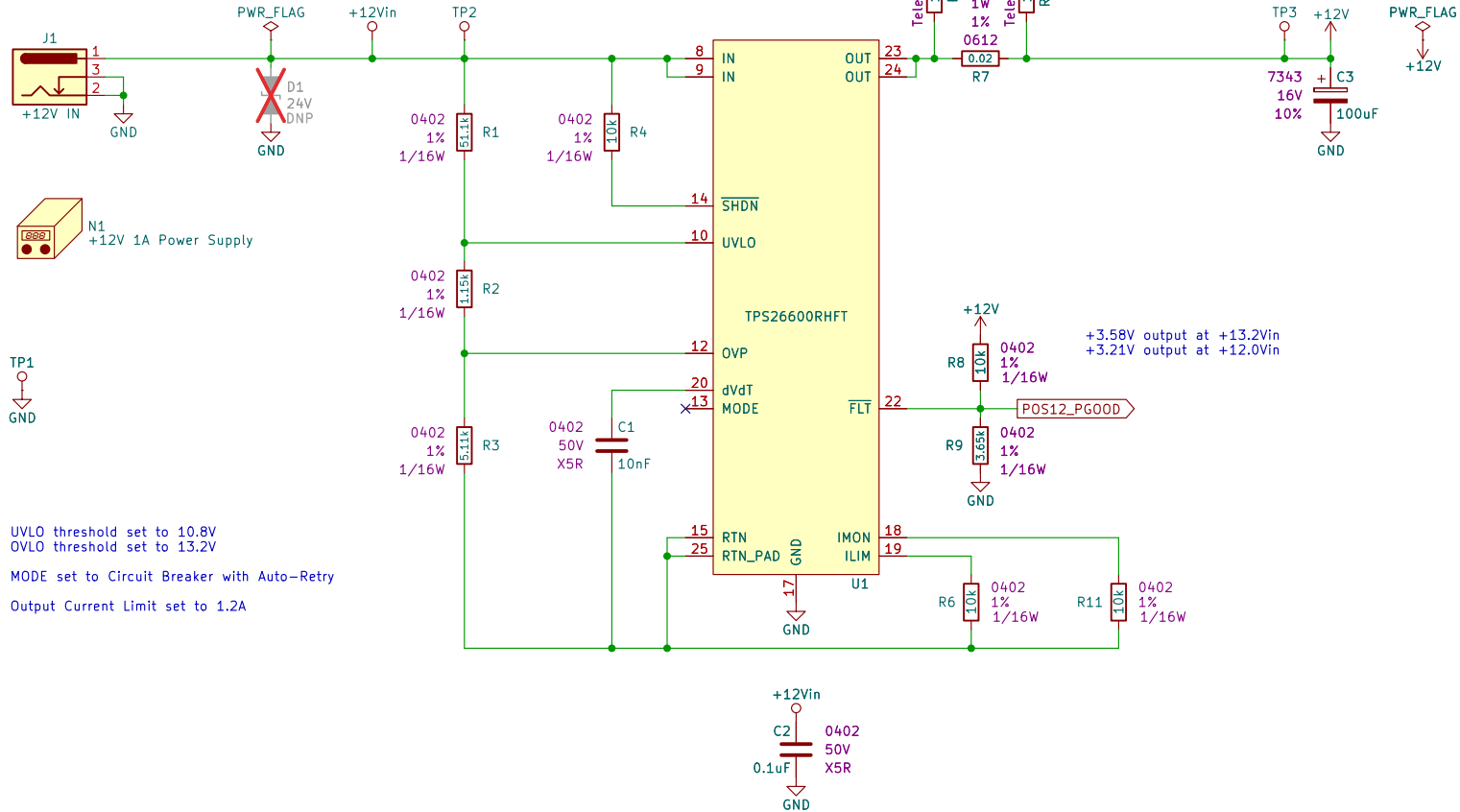


	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch	Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Buffers File: led_panel_buffers.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch				
B						
C						
D						
	1	2	3	4	5	

TRY TO TALLY LOAD CURRENTS

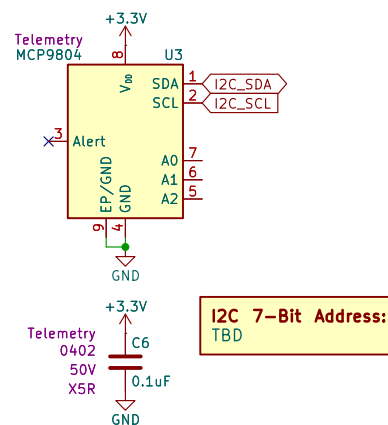
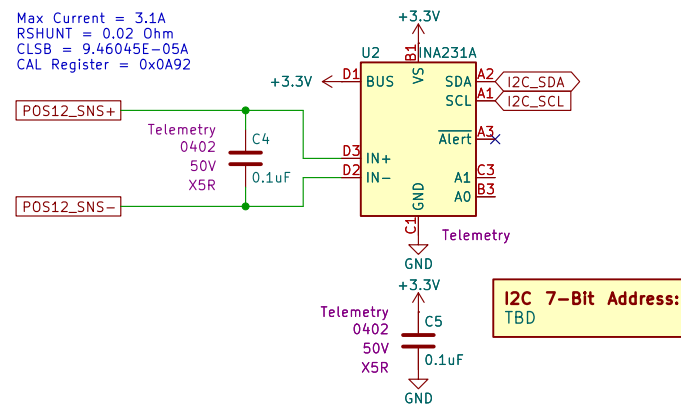


Sheet: /12V Power Input/
File: POS12_power_input.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

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Id: 2/20



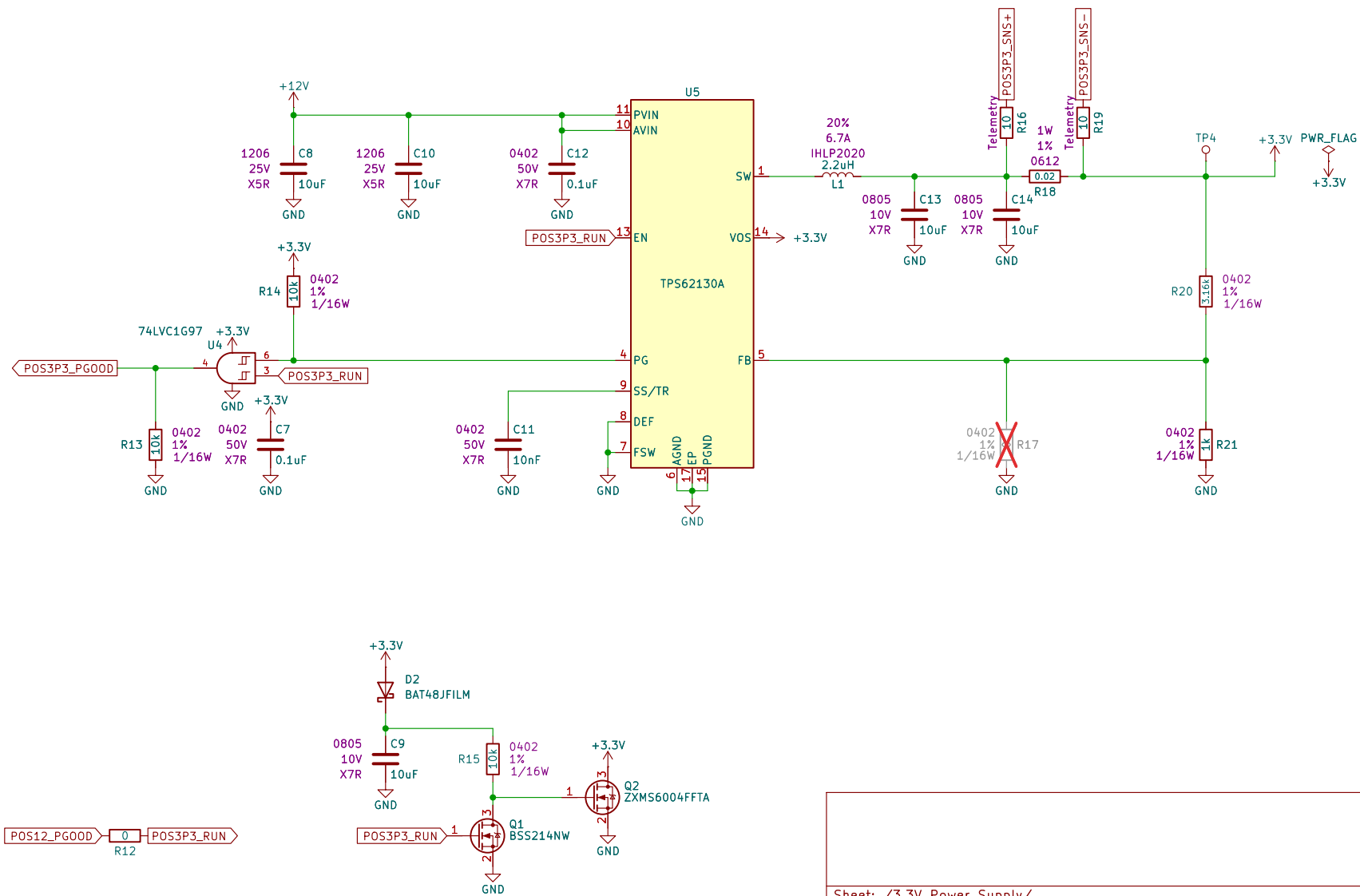
Sheet: /12V Telemetry/
 File: POS12_telemetry.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 3/20

TRY TO TALLY LOAD CURRENTS



Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

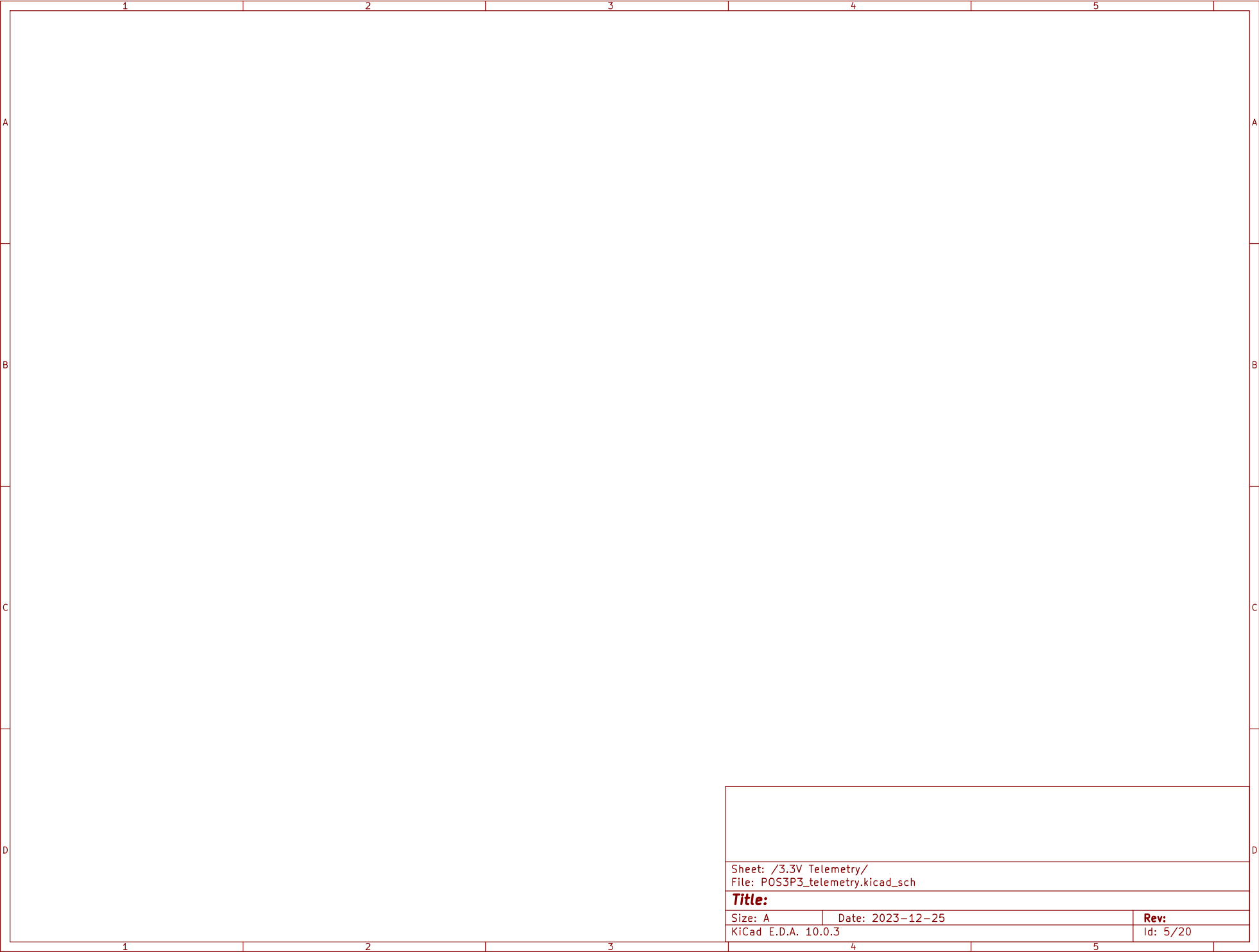
Title:

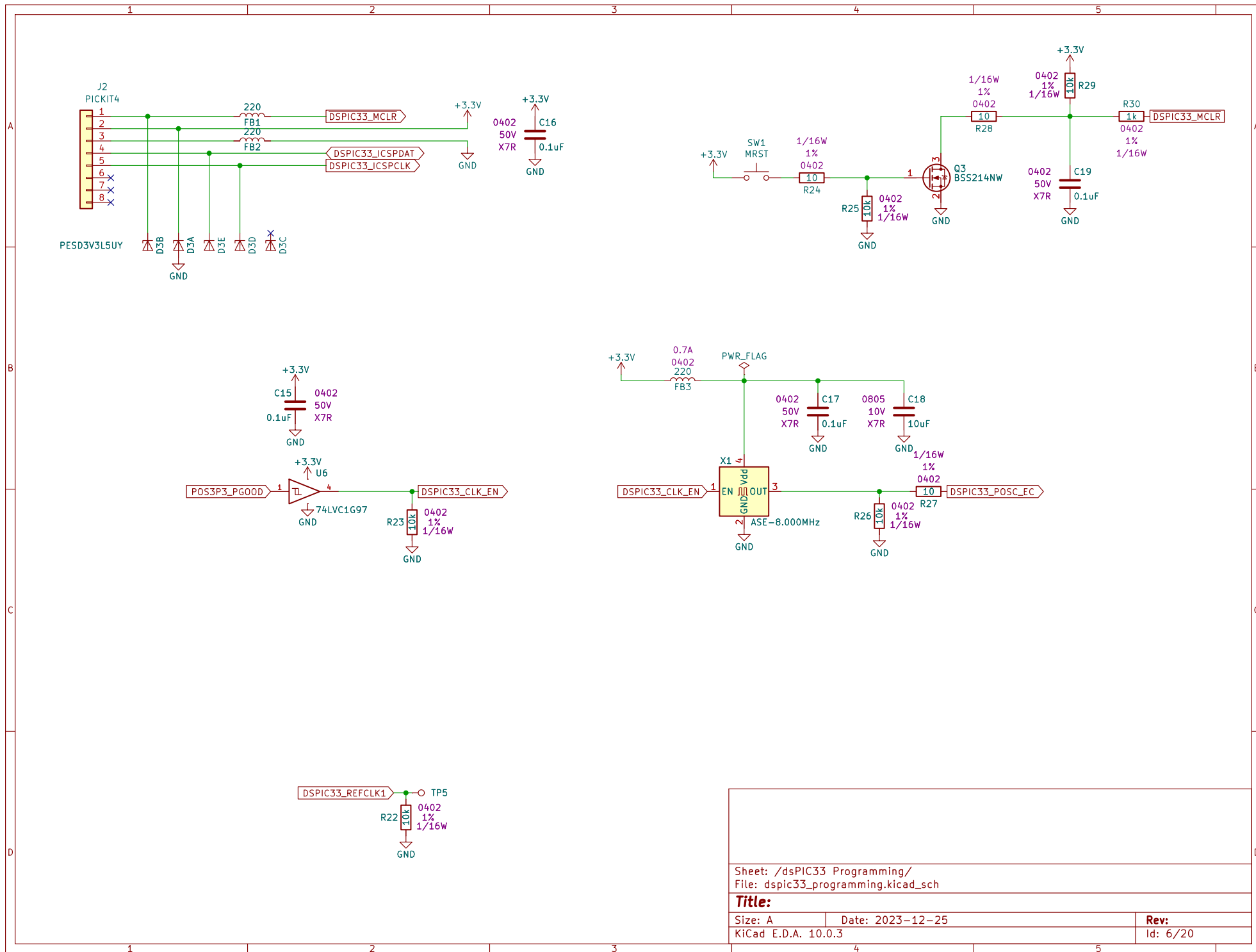
Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 4/20



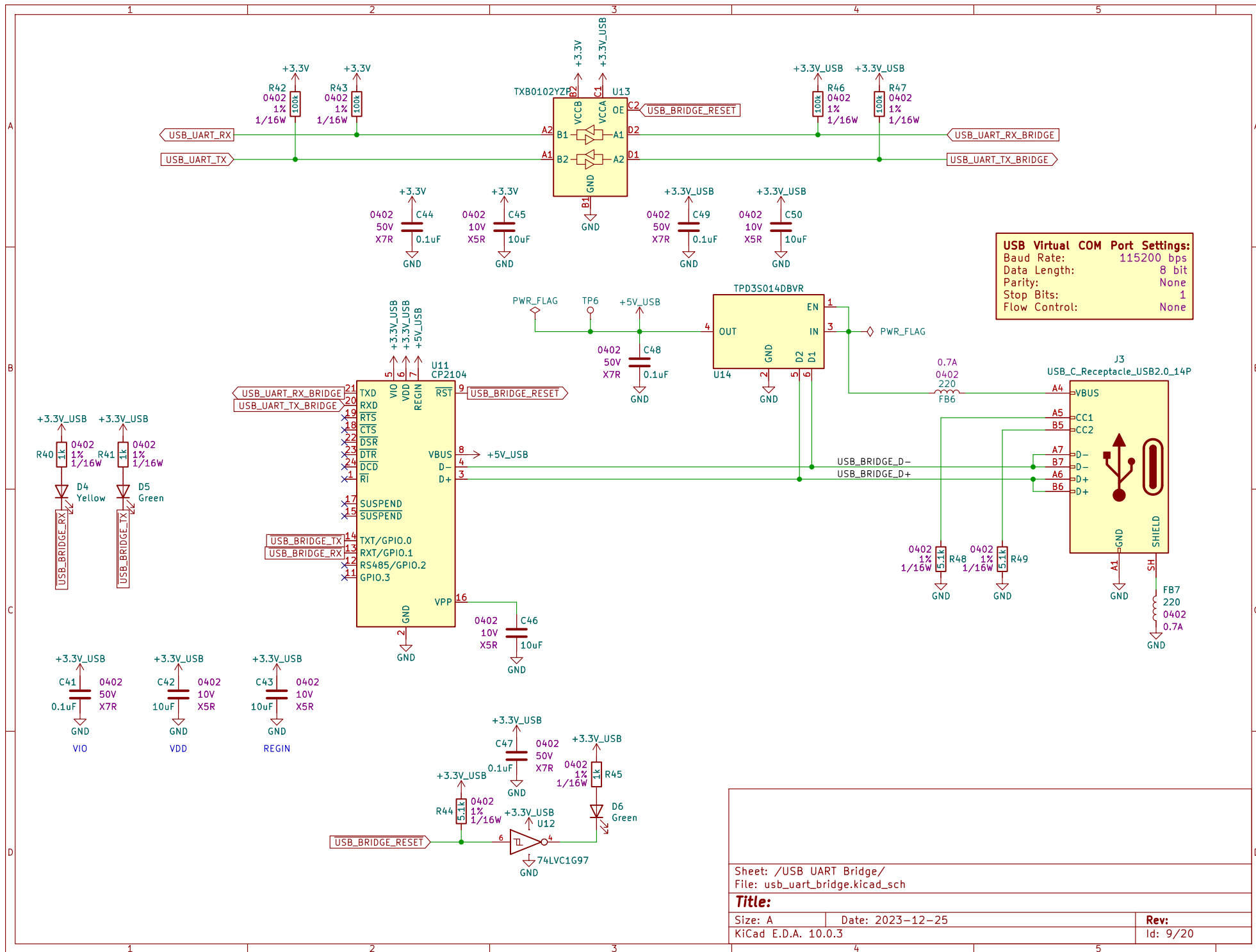


Sheet: /dsPIC33 Programming/
File: dspic33_programming.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 6/20

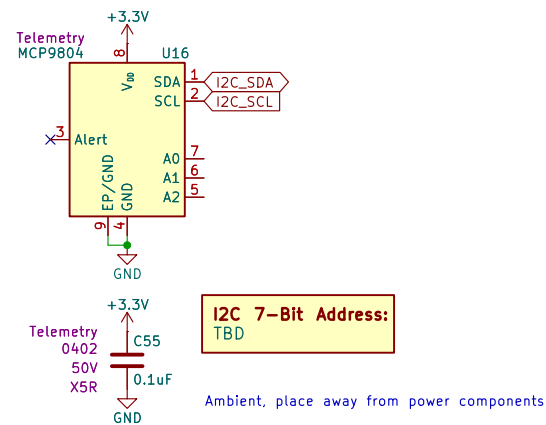
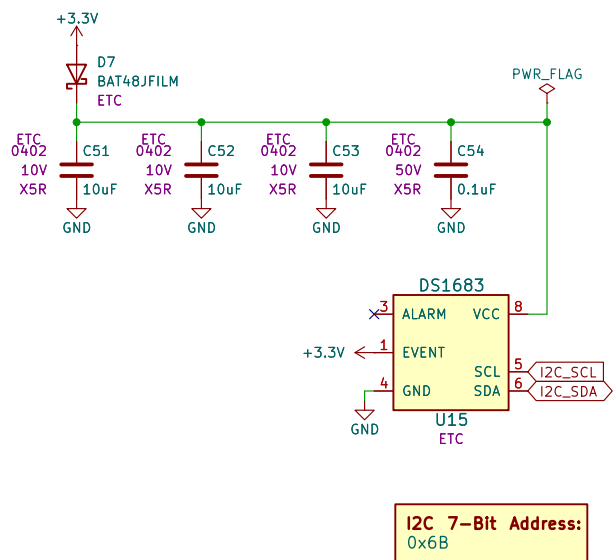


Sheet: /USB UART Bridge/
File: usb_uart_bridge.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 9/20

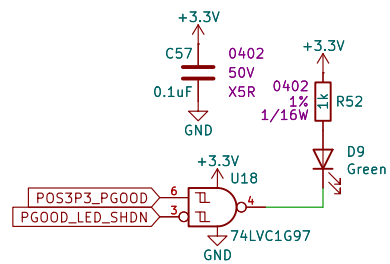
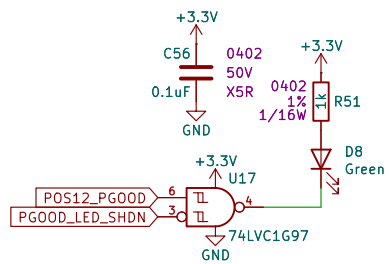
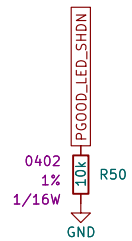


Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/20

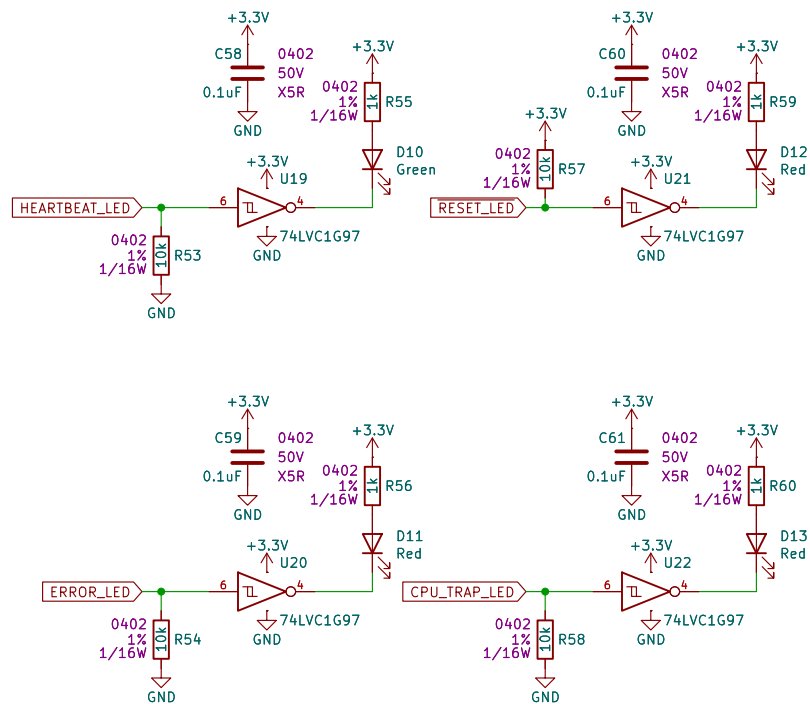


Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 11/20

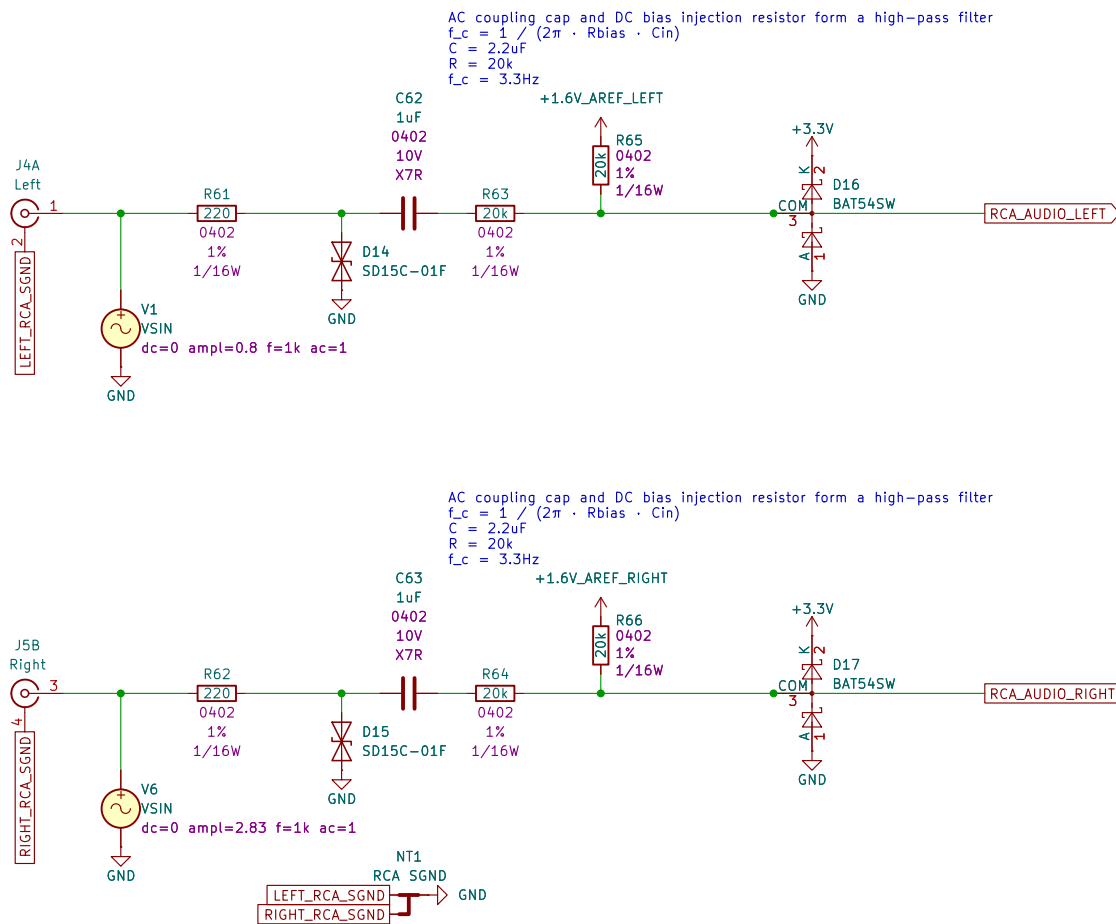


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/20



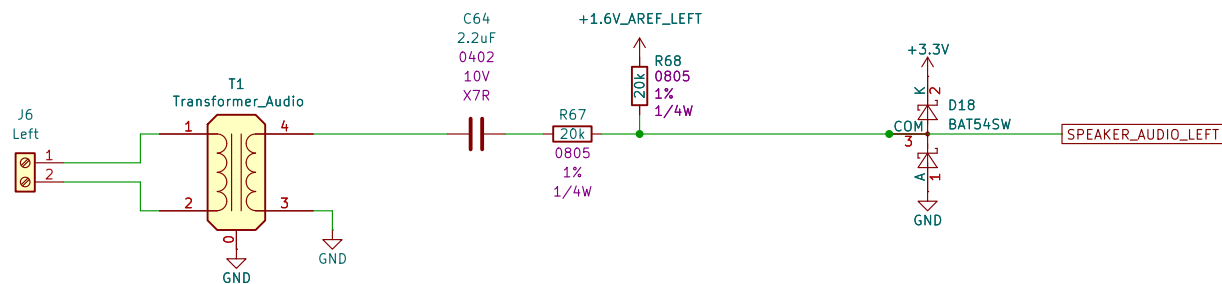
Sheet: /RCA Signal Conditioning/
 File: rca_signal_conditioning.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 13/20

Alternate is a 20:1 resistive attenuation (for 18V peaks) and a diff amp



Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

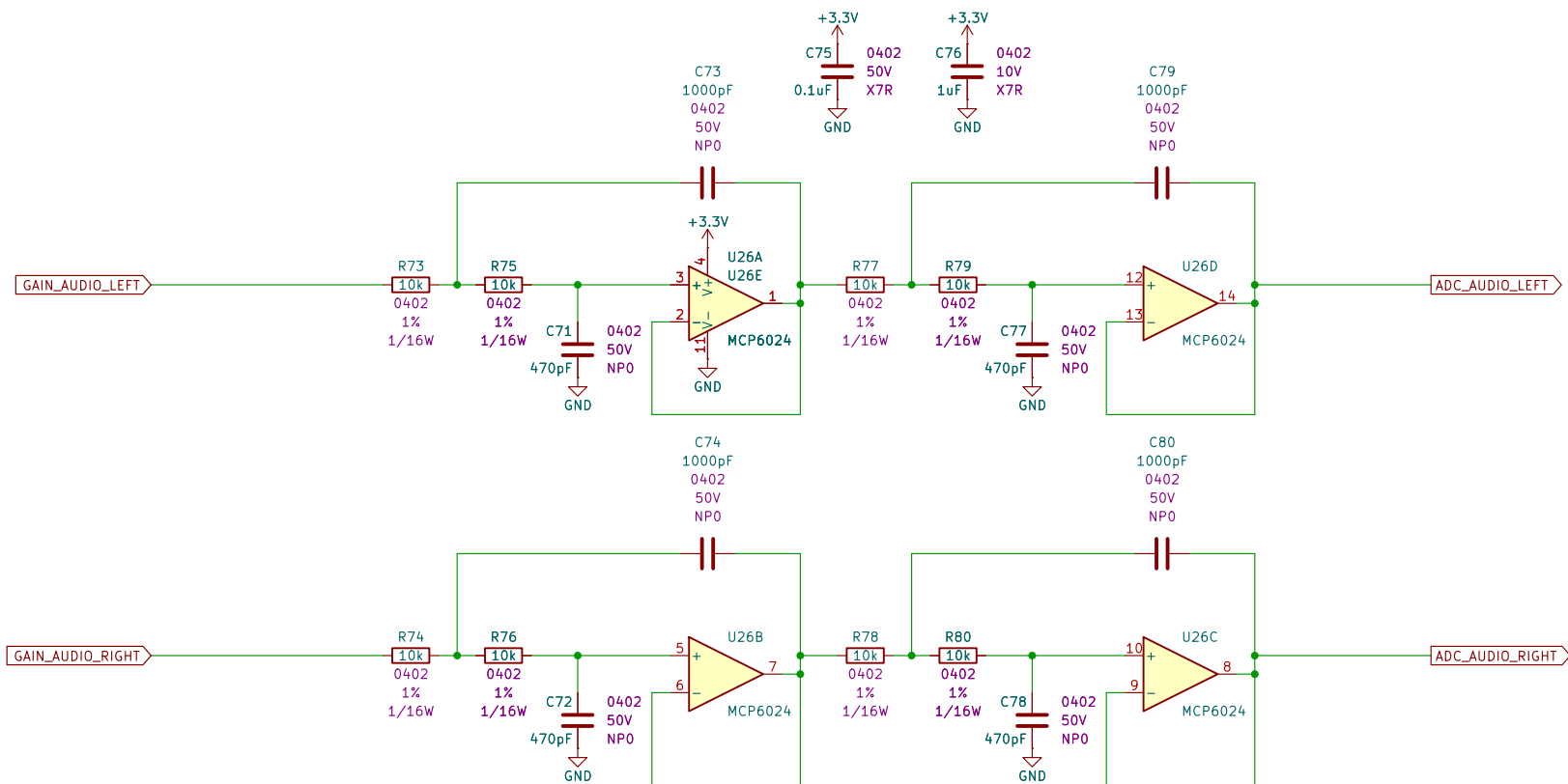
Title:

Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 14/20



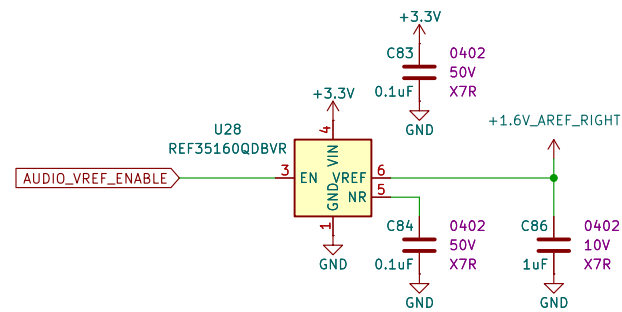
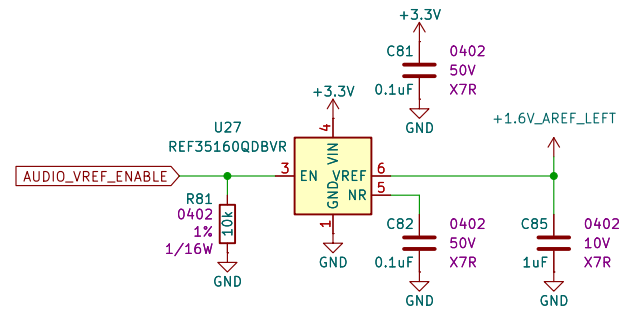
4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard around this frequency

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 16/20

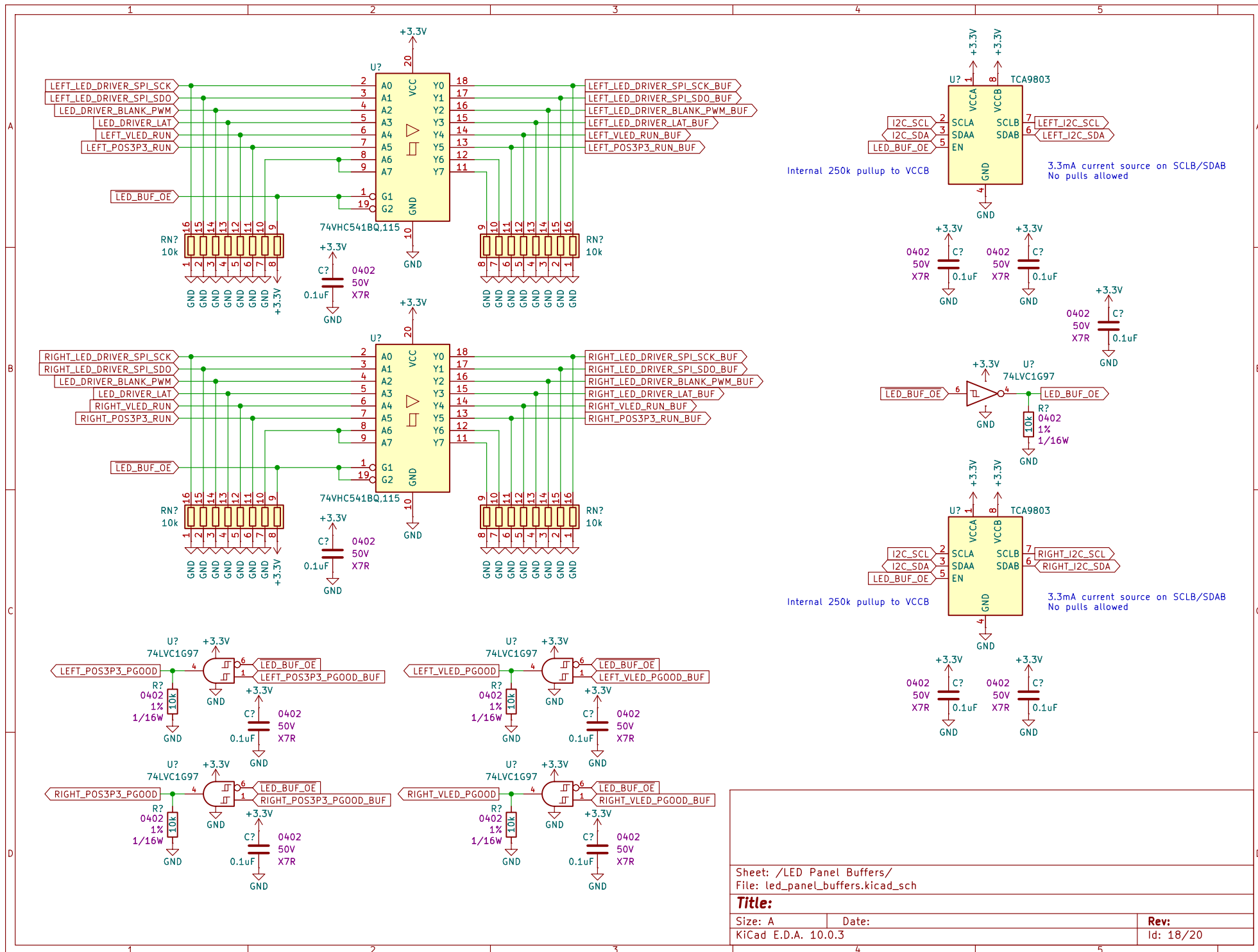


Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/20



J7
Power Button

SW2

J8
Power

SW?
RotaryEncoder_Switch

S1
S2
A
C
B

R84

+3.3V

U29

AT42071010-TSHR

5

3

4

6

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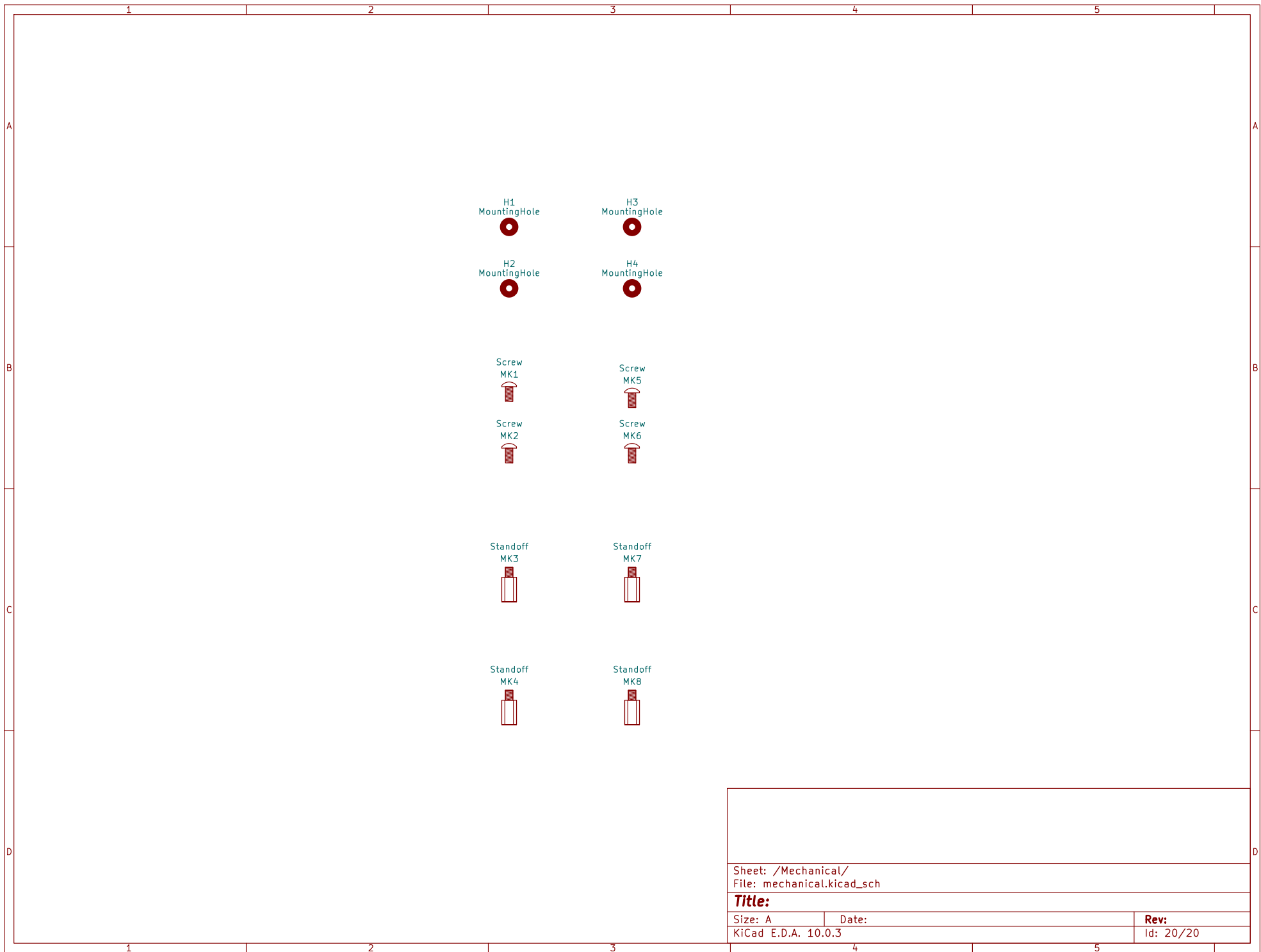
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H1
MountingHole

H3
MountingHole

H2
MountingHole

H4
MountingHole

Screw
MK1

Screw
MK5

Screw
MK2

Screw
MK6

Standoff
MK3

Standoff
MK7

Standoff
MK4

Standoff
MK8

Sheet: /Mechanical/
File: mechanical.kicad_sch

Title:

Size: A

Date:

Rev:

KiCad E.D.A. 10.0.3

Id: 20/20